A. Access Control

START

Create user accounts with usernames, passwords and roles

Set login status to FALSE

Display the login form

Ask the user for a username and password

IF Login is clicked THEN

IF username exists AND password is correct THEN

Set login status to TRUE

Save username and role

Refresh the application

ELSE

Display "Invalid username or password"

ENDIF

IF user is not logged in THEN

Stop the application

ENDIF

Create a page list containing Dashboard

Check the users role and add the pages allowed for that role

Display the allowed pages in the sidebar

IF Logout is clicked THEN

Set login status to FALSE

Clear username and role

Refresh the application

ENDIF

END

B. Worker Health and Safety Monitoring

START

Open Worker Safety page

IF CSV or Excel file is uploaded THEN

Read the file and add rows to worker records

ENDIF

Display form for worker ID, department, job role, shift, PPE, training, fatigue, near misses, previous incidents, methane, dust, temperature and safety observations

IF Analyse and Save is clicked THEN

IF Worker ID is empty THEN

Display "Please enter a Worker ID"

Create an empty warning list

IF PPE is No THEN add PPE warning. ENDIF

IF training is Incomplete THEN add training warning. ENDIF

IF fatigue is High THEN add fatigue warning. ENDIF

IF near misses > 0 THEN add near-miss warning. ENDIF

IF methane > 500 THEN add methane warning. ENDIF

IF dust > 5 THEN add dust warning. ENDIF

IF temperature > 40 THEN add temperature warning. ENDIF

IF (PPE is No AND fatigue is High) OR methane > 500

OR dust > 5 OR temperature > 40 THEN

Set risk to Critical.

ELSE IF warning count >= 2 OR methane > 300

OR dust > 3 OR temperature > 35 THEN

Set risk to High

ELSE IF warning count = 1 THEN

Set risk to Medium

ELSE

Set risk to Low

ENDIF

Save worker details, warnings and risk level

ENDIF

IF worker records exist THEN

Display worker table and CSV download option

ENDIF

END

C. Safety Incident Database

START

Open Safety Incidents page.

IF CSV or Excel file is uploaded THEN

TRY to read and save incident data

IF reading fails THEN display error. ENDIF

ENDIF

IF incident table is empty THEN

Display message saying there are no records

Ask for incident ID, type, severity and description

IF Add Incident is clicked THEN

Save the first incident and refresh application

ENDIF

ELSE

Show search box

IF search text is entered THEN

Search every column and show matching rows

ELSE

Show all incident rows

ENDIF

Display number of records shown

Find columns with fewer than 30 unique values

IF suitable columns exist THEN

Let user choose a column

Count each category and display bar chart

ENDIF

Offer CSV download of incident data.

ENDIF

END

D. Equipment Database and Condition Monitoring (D and E)

START

Open Equipment Monitoring page

IF CSV or Excel file is uploaded THEN

Read equipment file

IF Condition column is missing AND Temperature and Vibration exist THEN

FOR each row

IF temperature < 80 OR vibration < 5 THEN

OUTPUT “NORMAL: Equipment temperature approaching predefined threshold”

ELSE IF 80 <= temperature <= 100 OR 5 <= vibration <= 8 THEN

OUTPUT "WARNING: Equipment temperature approaching predefined threshold "
ELSE THEN

OUTPUT "CRITICAL: Equipment vibration exceeds predefined threshold. Inspection required"

ENDIF

Add imported rows to equipment records

ENDIF

Display form for equipment ID, type, manufacturer, operating hours, temperature, vibration, fuel, downtime, brakes, tyres, engine and maintenance status.

IF Save Equipment Record is clicked THEN

IF Equipment ID is empty THEN

Display error

ELSE

IF temperature > 100 OR vibration > 8 THEN

Set condition to CRITICAL

ELSE IF temperature >= 80 OR vibration >= 5 THEN

Set condition to WARNING

ELSE

Set condition to NORMAL

ENDIF

Set total time = operating hours + downtime

IF total time > 0 THEN

Set availability = operating hours / total time * 100

ELSE

Set availability = 100

ENDIF

Save record, display success message and refresh

ENDIF

IF equipment records exist THEN

Calculate missing availability values when possible

Display equipment table and CSV download

ENDIF

END

E. Equipment Maintenance Monitoring

START

Open Maintenance page

Get saved equipment records

IF equipment records exist THEN

Make a table of equipment records

Calculate missing availability values when columns exist

IF Maintenance status column exists THEN

Count maintenance categories and display bar chart

ELSE

Display missing-column message

ENDIF

IF Condition column exists THEN

Count equipment conditions and display bar chart

ELSE

Display missing-column message

Find equipment with availability < 85%

IF any exist THEN display low-availability table

ELSE display high-availability message. ENDIF

Find equipment with downtime > 15 hours

IF any exist THEN display high-downtime table

ELSE display no-excessive-downtime message. ENDIF

Display complete maintenance log

ELSE

Display "No equipment records found"

ENDIF

END

F. Risk Assessment

START

Open Risk Assessment page

Ask for activity or hazard description

Ask for likelihood (1 to 5) and consequence (1 to 5)

IF Assess Risk Score is clicked THEN

Set risk score = likelihood * consequence

IF score <= 4 THEN

Set classification to Low and give routine monitoring advice

ELSE IF score <= 9 THEN

Set classification to Medium and give caution advice

ELSE IF score <= 16 THEN

Set classification to High and give intervention advice

ELSE

Set classification to Critical and give stop-work advice

ENDIF

Save hazard, likelihood, consequence, score, class and feedback

Display feedback using the matching message style

ENDIF

IF risk records exist THEN display risk log. ENDIF

END

G1. Generate Active Alerts

START

Create an empty alert list

FOR each equipment record

Read equipment ID, temperature, vibration and maintenance status

IF temperature > 100 THEN display CRITICAL temperature alert.

ELSE IF temperature >= 80 THEN display WARNING temperature alert. ENDIF

IF vibration > 8 THEN display CRITICAL vibration alert

ELSE IF vibration >= 5 THEN display WARNING vibration alert. ENDIF

IF maintenance status is Overdue THEN display CRITICAL alert

ELSE IF maintenance status is Due THEN display WARNING alert. ENDIF

ENDFOR

FOR each worker record

Read worker ID, risk level, methane and dust

IF methane > 500 THEN add CRITICAL methane alert

ELSE IF methane > 300 THEN display WARNING methane alert. ENDIF

IF dust > 5 THEN add CRITICAL dust alert

ELSE IF dust > 3 THEN display WARNING dust alert. ENDIF

IF worker risk is Critical THEN display CRITICAL worker risk alert

ELSE IF worker risk is High THEN display WARNING worker risk alert. ENDIF

Return the alert list

END

G2. Display Alerts and Notifications

START

Open Alerts and Notifications page

Get all active alerts

IF active alerts exist THEN

Count CRITICAL and WARNING alerts

Display total alert count and alert table

Ask user to select a record ID

Find alerts matching the selected ID

FOR each matching alert

Display severity, category, ID, parameter, value,

status and recommended action

ENDFOR

ELSE

Display no-active-alerts message

Ask user to import or save worker orequipment records

ENDIF

END

H. Mining Monitoring Dashboard

START

Open Dashboard page

Get logged-in username

Choose a safety message using the username

Display welcome message and safety message

Allow user to choose dashboard card colour

Get equipment, worker and incident records

Count equipment records

Count worker records

Count all incident rows

Get all active alerts

Count alerts with CRITICAL severity

Display Equipment Monitored, Critical Alerts, Open Incidents and Workers Monitored cards.

Display welcome message showing the user's role.

END

I. Interactive Data Analysis

START

Open Analytics and Reports page

Show Equipment, Worker Safety and Engineering Questions tabs

IN Equipment tab:

IF equipment records exist THEN

IF Temperature and Vibration columns exist THEN

Display scatter chart, coloured by Condition if available

ENDIF

ELSE display no-equipment-data message. ENDIF

IN Worker Safety tab:

IF worker records exist THEN

IF Methane and Temperature columns exist THEN

Display scatter chart, coloured by Risk Level if available

ENDIF

ELSE display no-worker-data message. ENDIF

IN Engineering Questions tab:

IF equipment records exist THEN

Calculate maximum, minimum and average temperature

IF Availability column exists THEN

Calculate average and minimum availability

Calculate average downtime when column exists

Display equipment results

ELSE display equipment-data message. ENDIF

IF worker records exist THEN

Count workers whose PPE compliance is Yes

Calculate PPE compliance percentage

Calculate average and maximum methane where column exists

Display worker results

ELSE display worker-data message. ENDIF

END

J. Manage Users

START

Open Manage Users page

IF current role is not Administrator, THEN

Display "Access Denied"

ELSE

Display User Management heading

Create an empty user list

FOR each saved account

Get username and role

Add them to the user list

Display the user list in a table

ENDIF

END